Changes in Surface Water Bodies Associated With Madoi (China) Mw 7.3 Earthquake of May 21, 2021 Using Sentinel-1 Data

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***Abstract*— On May 21, 2021 (UTC), an Mw 7.3 earthquake occurred in Madoi County of Qinghai Province (China). There was no damage to buildings or loss of life; however, surface manifestation was observed in the epicentral areas. The analysis of synthetic aperture radar (SAR) and ground observation data shows changes in the section of the Yellow River and some lakes within the epicentral region associated with this strong earthquake. The changes in radar backscattering coefficient (BC) in water bodies are observed attributed to turbidity from the intense shaking of the earthquake. The liquefaction locations observed in the field within the epicentral region, which is also supported by the changes in the BC derived from Sentinel-1 data.**

***Index Terms*— 2021 Madoi earthquake, backscatter coefficient, satellite observation, synthetic aperture radar (SAR) data, Tibetan Plateau.**

I. Introduction

ON MAY 21, 2021 (18:04:13 UTC time), an earthquake (Mw 7.3) occurred in Madoi County, Guoluo Prefecture, Qinghai Province of China (hereinafter referred to as the Madoi M 7.3 earthquake). The epicenter [34.59°N, 98.34°E, source: China Earthquake Networks Center (CENC)] is located within the Bayan Har block, which is the upstream area of the Yellow River, with a large number of lakes and wetland areas. In the past 30 years, a few strong earthquakes (1997 M 7.5 Manyi, 2001 M 7.8 Kokoxili, 2008 M 7.9 Wenchuan, 2010 M 6.9 Yushu, 2013 M 6.6 Lushan, and 2017 M 6.5 Jiuzhaigou) occurred along the boundary of Bayan Har block (see Fig. 1). According to the earthquake catalog of CENC, as of May 30, 2021, nearly 3000 aftershocks have been recorded after the main earthquake event, and there were 63 aftershocks of magnitude M 3.0–3.9 and 13 of magnitude M 4.0–4.9, and the maximum magnitude of the aftershock was 5.1 that observed on May 22, 2021 [3]. These earthquakes occurred in remote places with altitudes 4200–4600 m where few people were living, which makes few losses of life and minor damages caused by this event. The focal mechanism of the main earthquake shows that the seismogenic fault (Kunlunshankou–Jiangcuo fault) is north-west (NW)-striking left-lateral strike-slip fault with normal faulting component and coseismic surface rupture of 160 km length located on the Jiangcuo segment [4], [5], which is consistent with the InSAR observation [6]. The largest coseismic left-lateral offset of 2.9 m was observed in the field [5]. The current slip rate of seismogenic fault is 1.2 ± 0.8 mm/year based on global positioning system (GPS) observations [7]. Liquefactions at different locations along the Yellow River were observed after the main event [4], [5], [8].

The synthetic aperture radar (SAR) backscatter signal of small water bodies is lower compared to other land cover, being smooth surface, which facilitates the detection of water bodies. In larger water bodies, water waves increase the backscattering response due to surface roughness, whereas the backscattering coefficient (BC) is not affected in small water bodies. Sentinel-1 satellite data have been used to detect flood [9]–[11] and monitor surface water bodies [12], [13].

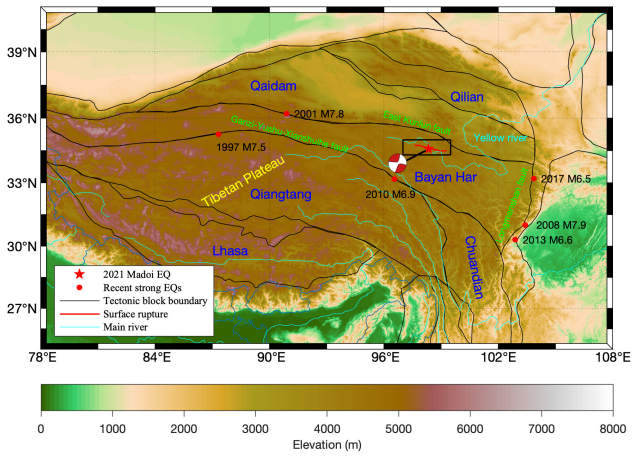


Fig. 1. Epicenter location (source: CENC) and focal mechanism solution [source: Global Centroid Moment Tensor (GCMT)] of the Madoi M 7.3 earthquake of May 21, 2021. The base map is made using ETOPO1 1 arc-minute gridded elevation data [1]. The boundaries of active-tectonic block data are taken from [2]. The coseismic surface rupture is taken from [4]. Black box area shown in this figure is considered in Fig. 3(a).

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Recently, BC from Sentinel-l combined with satellite optical data has been used to study the changes in the flood plains and water quality associated with the Chamoli disaster [14]–[16]. The suspended solids concentration in the surface water bodies is directly related to the SAR BC [17]. Within the epicentral region, the increasing turbidity (with high suspended solids concentration) in water bodies, including river and lake water, shows the contrasting changes in BC under the influence of seismic wave propagation.

In the present study, combined multisatellite and ground observation data have been used to study the changes in the water bodies within the epicentral region. We first defined BC residual and obtained the BC changes in the water bodies of Yellow River and 83 lakes during April–June 2021. The possible contributing factors, such as precipitation, water/ice cover, surface roughness, changes in dielectric constant, and liquefaction, have been analyzed one by one to find the causes of high BC residual. Finally, four-year (2017–2020) BC changes during the same period were analyzed to compare the distinct BC changes associated with the 2021 Madoi M 7.3 earthquake.

II. Tectonic, Seismic, Meteorological, and Hydrological Features

The Madoi M 7.3 earthquake occurred in the Bayan Har tectonic block, which is located in the northern Tibetan Plateau (see Fig. 1), with the strongest seismic activity in China. The movement of Bayan Har block is in the direction of northeast (NE) 62° and with a rate of ∼21 mm/year based on the GPS observation [2]. The magnetotelluric profile indicates the source area between high conductivity and high resistivity blocks in the middle and lower crust [18]. The northern boundary of Bayan Har tectonic block is the East Kunlun fault zone, striking NW-west (NWW) left-lateral strike-slip fault. The 2001 M 7.8 Kokoxili earthquake was the largest earthquake in this fault zone. The southern boundary is the NW-striking Ganzi–Yushu–Xianshuihe fault zone (left-lateral strike-slip fault) where two strong earthquakes (1997 M 7.5 Manyi and 2010 M 6.9 Yushu earthquakes) occurred in the past three decades. The eastern boundary is the NE-striking right-lateral strike-slip Longmenshan fault zone, where the deadly 2008 M 7.9 Wenchuan earthquake occurred. Fig. 1 shows the spatial distribution of elevation and tectonic blocks in the Tibetan Plateau. Fig. 2 shows the daily seismic frequency and magnitude recorded within a 500 km radius in the epicentral region during January 2020–August 2021 [United States Geological Survey (USGS)]. It clearly shows earthquakes with $M > 4.0$ prior to the main earthquake event of May 21, 2021 in this region since 2020. Within a 50 km radius, no earthquakes with $M > 6.0$ were recorded. The last biggest earthquake (magnitude 6.0) occurred on March 24, 1971 within a 100 km radius around the epicenter of the Madoi M 7.3 earthquake.

The epicentral region is located close to the source area of the Yellow River, controlled by the alpine climate (cold and dry conditions) [19]. According to the long-term observation data (1957–2016) from the Madoi meteorological station, the multiyear monthly averaged highest temperature is about 6.8 °C in the month of July and the lowest −16.8 °C in the month of January [20]. More than 80% of the precipitation occurs during June–August [21]. Within the epicentral region, a large number of lakes exist, which were fed by precipitation, snow/ice melting, and river water [22]. A hydraulic connection exists among rivers, lakes, and groundwater underlain by seasonal frozen ground [23].

TABLE I
Sentinel-1 Data Used in the Present Study

Satellite	Acquire Time	Flight Direction	Incidence angle
Sentinel-1A	2021-04-02 23:28:19	Descending	39.54
Sentinel-1A	2021-04-14 23:28:19	Descending	39.54
Sentinel-1A	2021-04-26 23:28:20	Descending	39.54
Sentinel-1A	2021-05-08 23:28:21	Descending	39.54
Sentinel-1A	2021-05-20 23:28:21	Descending	39.54
Sentinel-1B	2021-05-26 23:27:26	Descending	39.48
Sentinel-1A	2021-06-01 23:28:22	Descending	39.54
Sentinel-1A	2021-06-13 23:28:23	Descending	39.54
Sentinel-1B	2021-06-19 23:27:40	Descending	39.49
Sentinel-1A	2021-06-25 23:28:23	Descending	39.54

III. Data Used

A. Satellite Observation Data

The European Space Agency (ESA) Sentinel-1 mission is a two-satellite constellation (Sentinel-1A and -1B) to obtain 6-day (two satellites) or 12-day (one satellite) revisit time C-band (centered at 5.405 GHz) SAR observation data. In the present study, we have used Sentinel-1 interferometric wide (IW) swath mode data with a spatial resolution of 20 m × 5 m at vertical transmit and vertical receive (VV) polarization due to the better sensitivity to water body compared to vertical transmit and horizontal receive (VH) polarization [24]. The radiometric accuracy standard deviations of IW is 0.30 dB for Sentinel-1A [25] and 0.29 dB for Sentinel-1B [26]. Sentinel-1 Level-1 ground range detected (GRD) high resolution (HR) IW data have been chosen in the present study, which is downloaded through the Alaska Satellite Facility (ASF, https://asf.alaska.edu/). We computed BC at VV polarization using the Sentinel Application Platform (SNAP) software developed by ESA (https://step.esa.int/main/toolboxes/snap/). The preprocessing steps include thermal noise removal, calibration (dB), speckle filtering, and terrain correction. Table I gives the details of Sentinel-1 data covering the study area during April–June 2021.

Chinese Gaofen-1 (GF-1) and Gaofen-2 (GF-2) satellite data, with a spatial resolution of 2 and 1 m, respectively, are used to study the BC changes in water bodies. Cloud-free GF-1 image on May 3, 2021 (as preevent) and GF-2 image on June 5, 2021 (as postevent) are acquired from the China Center for Resources Satellite Date and Application (CRESDA).

Precipitation and soil moisture data have also been used to analyze the observed changes in BC. The National Aeronautics and Space Administration (NASA) Global Precipitation Measurement (GPM) Integrated MultisatellitE Retrievals for GPM (IMERG) precipitation product with a spatial resolution

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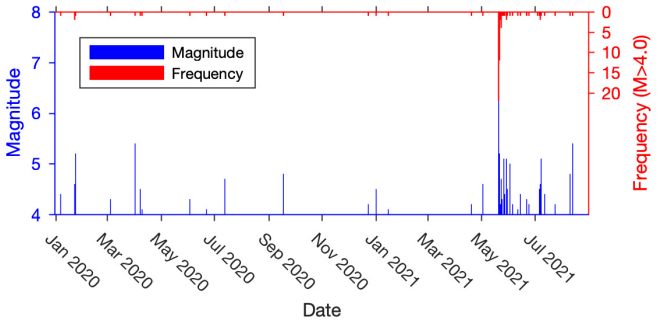


Fig. 2. Daily seismic frequency and magnitude observed within 500 km radius around the epicenter of Madoi M 7.3 earthquake during January 2020–August 2021. A number of earthquakes with $M > 4.0$ occurred during one and a half year prior to the main earthquake.

of 0.1° × 0.1° [27] and the Advanced Microwave Scanning Radiometer 2 (AMSR2) soil moisture data with a spatial resolution of 10 km are used in the present study. The daily soil moisture has been computed from the ascending (daytime) and descending (nighttime) daily products to avoid any orbital gap in data. The 0.25° daily NASA Global Land Data Assimilation System (GLDAS) soil moisture data has been used as well.

B. Ground Observation Data

The meteorological parameters (temperature and precipitation) are important to understand the backscattering behavior. The daily meteorological data from the closest meteorological station to the epicenter—Madoi station (location: 34.91°N, 98.21°E, elevation: 4273 m)—are downloaded through the National Oceanic and Atmospheric Administration (NOAA) Global Historical Climatology Network (GHCN, https://gis.ncdc.noaa.gov/maps/ncei/cdo/daily) [28].

IV. Results and Analysis

A. Variability in BCs in Water Bodies

The 2021 Madoi M 7.3 earthquake occurred in the upstream area of the Yellow River. The nearest distance from the epicenter (CENC) to the mainstream of the Yellow River is less than 2 km. The BC changes in the water bodies (Yellow River and lakes) have been studied using Sentinel-1 data. The BC value on May 20, 2021 (the closest day prior to the day of main earthquake event on all available Sentinel-1 data as shown in Table I) is selected as a reference. The residual BC values were computed using the BC value on the other available days minus the reference BC value.

1) Mainstream of the Yellow River: We obtained the Sentinel-1 derived BC residual of postevent days (May 26, June 1, 13, and 19, 2021) to study the changes in the surface BCs. Fig. 3(a) shows the spatial variations of BC residual value on May 26, 2021 (five days after the main earthquake event). Along the mainstream of the Yellow River, two segments with obvious increased BC have been found [in Fig. 3(a)]. The red arrows in Fig. 3(b) and (c) show the increased BC along the Yellow River. Region A covers the epicentral region of the Madoi M 7.3 event and region B is located about 30-km southeast of the epicenter. Compared with the other segments of the Yellow River, the two segments are with the closest distance to surface rupture zone that observed by the field investigation [4] [red line in Fig. 3(a)].

It can be seen from the regional enlarged map for BC changes in Box A on May 26, 2021 [see Fig. 3(b)], the high BC positive residual values are distributed along the Yellow River and close to the Yellow River town [blue dot in Fig. 3(a)], where collapsed house and broken bridge have been found after the Madoi M 7.3 earthquake [8]. However, in the same river segment on June 1, 13, and 19, 2021, the negative BC residual values have been observed [see Fig. 3(d), (f), and (h)]. Besides changes along the river, we also observed the strong increase in backscattering signals on land surface within the epicentral region on June 1, 2021 [see Fig. 3(d)] and other locations in the whole area of Box A on June 19, 2021 [see Fig. 3(h)]. The reason for the increased BS on land surface will be analyzed in Section IV-B.

In Box B, the similar BC variations were observed [see Fig. 3(c), (e), (g), and (i)]. The high BC positive residual was distributed along the mainstream of the Yellow River on May 26, 2021 [red arrows in Fig. 3(c)]. On June 1, 13, and 19, a large number of high BC positive residuals have been found on the land surface. The surface BC mainly depends on surface roughness, moisture content, and so on, and the observed BC changes in land surface could be related to roughness and soil moisture in such a short period.

2) Lakes: A total of 83 lakes (not less than 100 m wide) in the epicentral and surrounding region were selected to study the BC changes in lakes. The BC values derived from Sentinel-1 averaged over 100 m × 100 m area in lakes have been computed. Similar to the method used in studying the Yellow River, we computed the BS residual values on the available preevent and postevent days using the data on May 20, 2021 (one day prior to the main earthquake) as the reference. Five BC residual ranks (>5, 4–5, 3–4, 0–3, and <0 dB) have been defined. The red squares in Fig. 4 indicate those lakes with high BC positive residual value (greater than 5 dB). The distinct increased BC on April 14 and 26 have been observed in many lakes [see Fig. 4(a) and (b)]. A total of 51 and 29 out of 83 lakes show high BC positive residual (greater than 5 dB) on April 14 and 26, 2021, respectively. These lakes are scattered throughout the whole region from west to east. The total of lakes with high BC positive residual reduced to 8 and 14 on May 8 and 26, 2021, respectively [see Fig. 4(c) and (d)]. It is important to note that the distribution of the lakes with high positive residual on May 26, 2021 [see Fig. 4(d)] seems to depend on the location compared with those on May 8, 2021. The 14 lakes with high BC positive residual on May 26, 2021 are located close to the epicenter and distributed along the coseismic surface rupture zone (red line in Fig. 4) compared to the preevent days [see Fig. 4(a)–(c)]. After that, the number of lakes with high positive residual in BC declines on June 1, 13, and 19, 2021 [see Fig. 4(e)–(g)].

B. Diagnostic Analysis of Contributing Factors

In general, the changes in BC could be related to various factors, such as precipitation, water/ice cover, change in turbidity of water, roughness of surface, and changes in dielectric

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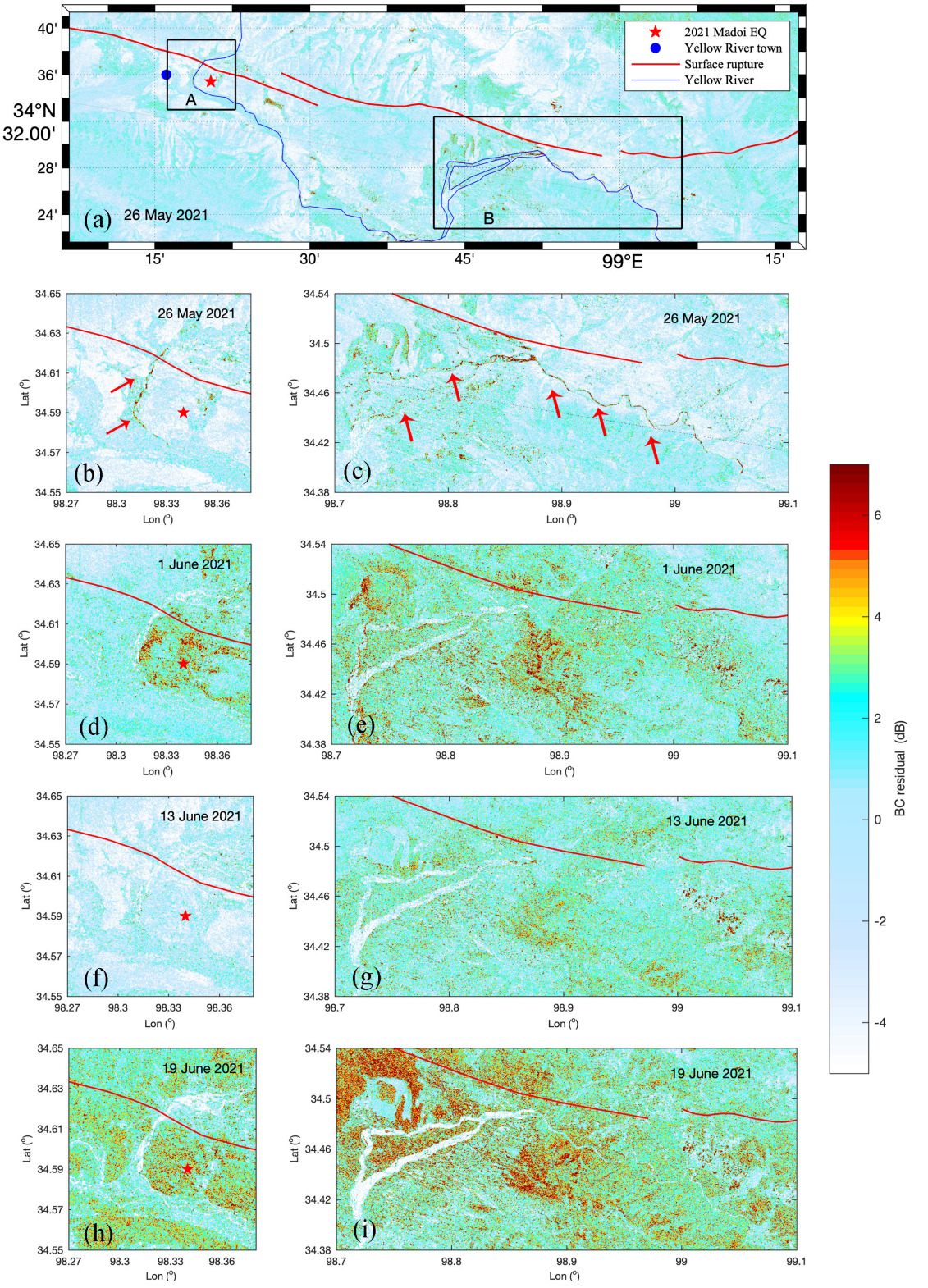


Fig. 3. Spatial distribution of BC residual on the available days around the occurrence of Madoi M 7.3 earthquake using the reference image of May 20, 2021. (a) Locations with changes in BC on May 26, 2021. (b), (d), (f), and (h) BC residual distribution in Box A on May 26 and June 1, 13, and 19, 2021, respectively. (c), (e), (g), and (i) BC residual distribution in Box B on May 26 and June 1, 13, and 19, 2021, respectively.

constant associated with the different types of surfaces. Here, we analyzed these factors one by one to find out the main source or the cause for the BC changes in water bodies around the epicenter of 2021 Modoi earthquake.

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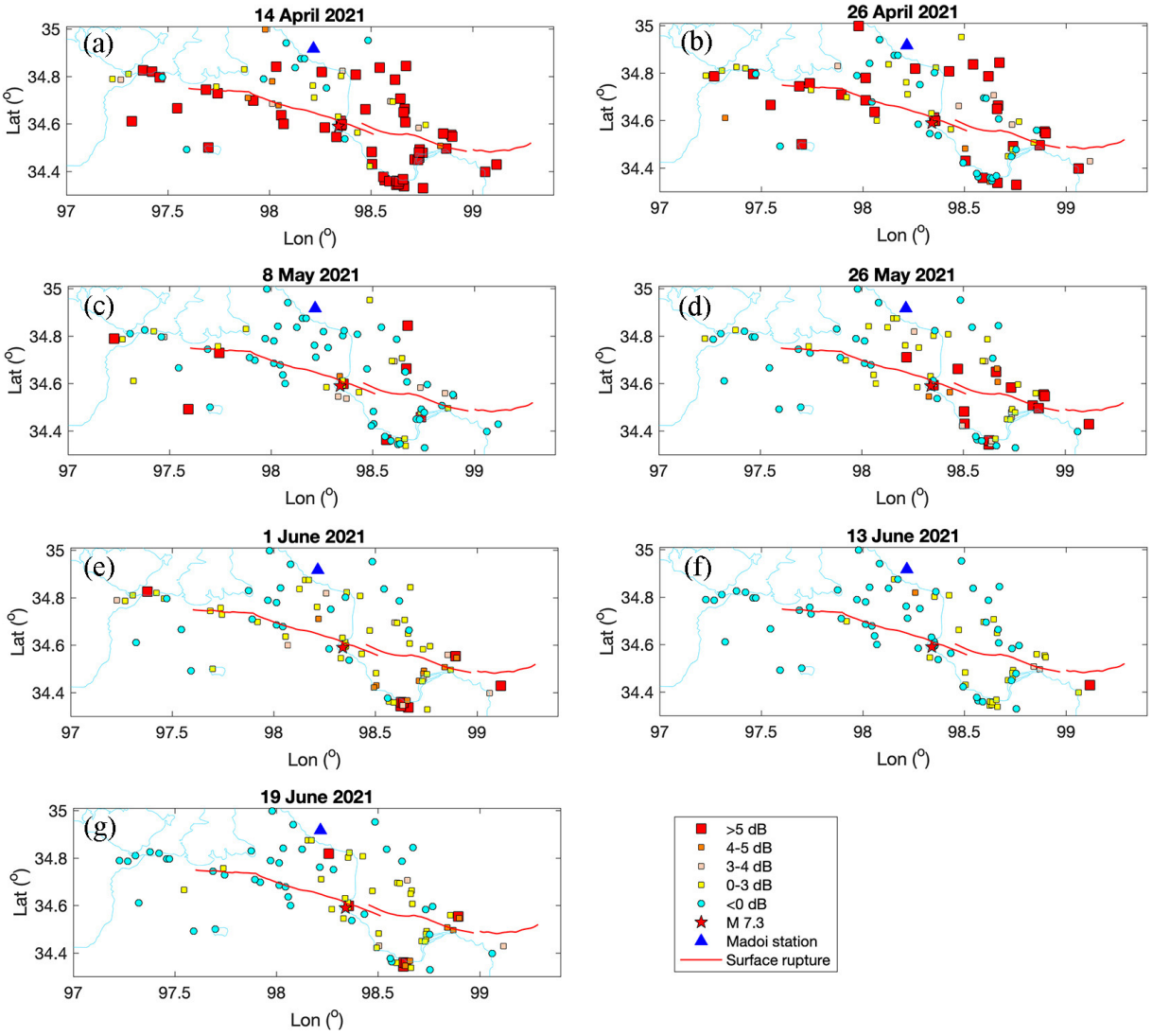


Fig. 4. BC residual ranks on the 83 selected lakes on the available days around the occurrence of Madoi M 7.3 earthquake. (a)–(g) April 14 and 26, May 8 and 26, and June 1, 13, and 19, 2021, respectively.

1) Precipitation: The results of simulated rain experiment show that the effect on the SAR backscatter from the water surface is significant when the raindrops are larger at low wind speed [29]. The effects of rain on SAR backscatter from the ocean surface have been studied widely [30], [31]. We obtained the precipitation data from satellite and ground station to study the BC variations on the water bodies, although this effect is almost impossible to occur on the surface of river and lake. Fig. 5 shows the *in situ* precipitation data of Madoi station (green bar in Fig. 5) and GPM precipitation data (red bar in Fig. 5). We can see that no significant rainfall events have been observed on May 26, 2021 on both, the site record from Madoi station and the area-averaged (34.38°–34.65°N, 98.27°–99.1°E, covered) precipitation from the GPM product taken from NASA Giovanni portal (https://giovanni.gsfc.nasa.gov/giovanni/). The increased BC residuals on May 26 in the Yellow River and the lakes [see Figs. 3(b) and (c) and 4(d)] are not caused by the heavy rainfall event.

Precipitation can increase soil moisture on the land surface, which increases BC values. In this regard, we analyzed the variations of soil moisture in the epicentral region because it can cause changes in BC on the land surface. The used region of AMSR2 soil moisture data is the same as GPM precipitation data. The soil moisture data and precipitation data show a close relationship (see Fig. 5), although we cannot get soil moisture data at high spatial resolution to relate one-to-one BC and soil moisture values. The increased soil moisture has been observed during June 16–23, 2021, which is related to the precipitation. Furthermore, we tried to find out whether the changes in BC on the land, shown in Fig. 3(d)–(i), are related to the increased soil moisture due to precipitation. The low soil moisture and precipitation values were observed on May 26 and June 1, and also on earlier days, a slight increase in

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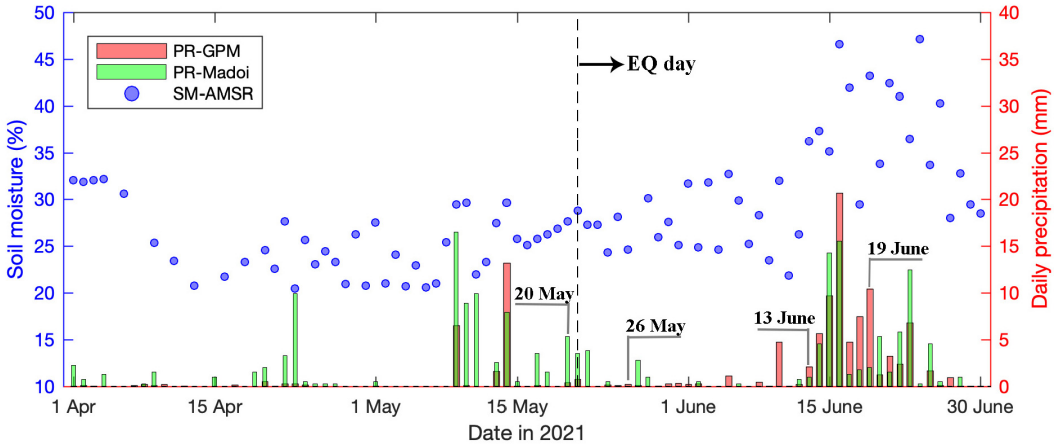


Fig. 5. Daily precipitation and soil moisture observed by satellite and ground station during April–June 2021. SM-AMSR stands for area-averaged (34.38°–34.65°N, 98.27°–99.1°E) daily soil moisture observed by AMSR2. PR-GPM stands for GPM daily precipitation in the same region as SM-AMSR. PR-Madoi stands for precipitation observed at Madoi station. Vertical black dashed line shows the earthquake day.

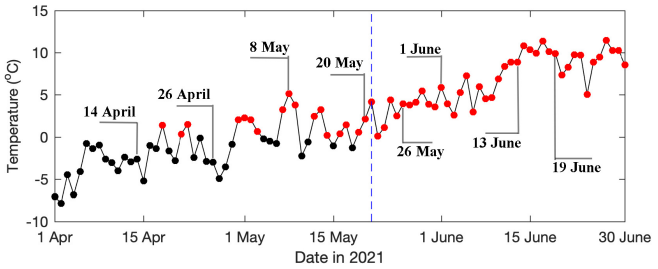


Fig. 6. Daily averaged temperature observed in Madoi meteorological station during April–June 2021. Red dots show the days with averaged temperature greater than 0 °C. Vertical blue dashed line shows the earthquake day.

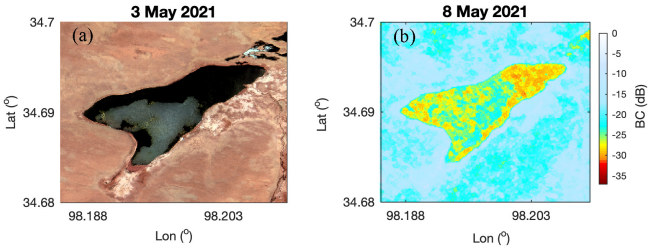


Fig. 7. Distinct difference between lake water and lake ice. (a) GF-1 true-color image with a spatial resolution of 2 m on May 3, 2021. (b) Sentinel-1-derived BC (VV polarization) on May 8, 2021.

BC was observed on May 26 [see Fig. 3(b) and (c)] and large increase on June 1 [see Fig. 3(d) and (e)] is not caused by the soil moisture changes. On June 13, 2021, the slight increase in BC value was observed in Box B [see Fig. 3(g)], which could be associated with the increased soil moisture due to the small precipitation (retrieved from GPM data over the study area) on June 10, 2021. No precipitation has been recorded during this period on the Madoi meteorological station (point data) (close to Box A, about 30 km), which could be the reason for no enhancement in BC residual value on June 13, 2021 in Box A [see Fig. 3(f)]. As for the distinct enhanced BC in both on June 19 [see Fig. 3(h) and (i)], this could be attributed to the increased soil moisture caused by the precipitation during this period. The permafrost thaw likely enhances soil moisture [32] that could increase a BC value on land as well.

2) Ice/Water Cover: The SAR C-band BC (in VV polarization) is significantly different between liquid and solid states of water due to their different dielectric properties. BC of lake ice is greater than the calm lake water [33]. We observed a number of lakes with high BC residue on April 14 and 26, 2021 [see Fig. 4(a) and (b)]. From the daily temperature variations during April–June 2021 observed from the Madoi meteorological station (see Fig. 6), the temperature was found to be −2.6 °C and −3 °C on April 14 and 26, respectively. Under this condition, the lake water is likely to be under frozen state showing high BC values. The temperature above 0 °C observed on April 18, 21, and 22, 2021 induce to the melting of lake ice, which reduces the number of frozen lakes on April 26, 2021 compared to April 14, 2021 showing changes in BCs [see Fig. 4(a) and (b)]. A higher temperature with 5.1 °C on May 8, 2021 could have caused melting of ice-covered lakes. The true-color image from the GF-1 satellite with a spatial resolution of 2 m on May 3, 2021 and Sentinel-1 BC on May 8, 2021 for the same lake clearly shows the changes in BC due to the melting of lake ice (see Fig. 7). With the increase in temperature, only few lakes with the high BC residual value were observed on May 8, 2021 [see Fig. 4(c)]. However, on May 26, 2021, the ice was almost impossible to exist on the lake surface since the temperature reached 3.9 °C and the temperature above 0 °C lasted for many days prior to this date. Therefore, the main factor for the increased number of lakes with high BC positive residual on May 26, 2021 [see Fig. 4(d)] is not related to the state of lake water.

3) Roughness of Surface: The BC changes are also related to the roughness of surface. For the water surface, the wave that induced by strong wind can change the roughness. We analyzed the mean and maximum wind speed data during the months from April to June 2021 from the Madoi meteorological station (see Fig. 8), and no strong wind was recorded during this period. For land surface, the change in

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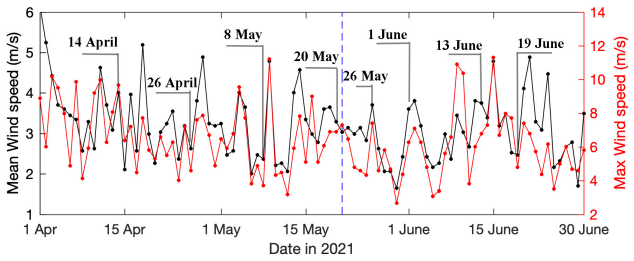


Fig. 8. Daily wind speed observed in Madoi meteorological station during April–June 2021. Black line indicates mean wind speed and red line shows maximum wind speed. Vertical blue dashed line shows the earthquake day.

vegetation cover can affect the roughness. However, the huge change in vegetation is unlikely to occur in the short time. The BC changes observed here may not be due to the changes in surface roughness.

4) Other Factors: The dielectric constant of water bodies changes with temperature. We did not obtain the water temperature of the Yellow River and the lakes in the epicentral region. However, in general, the temperature of open water is related to the air temperature on the same day. From the temperature observed in Madoi meteorological station (see Fig. 6), we can obtain that the temperature difference is 1.8 °C–7.8 °C from the reference day (May 20, 2021) and the available postevent days (May 26 and June 1, 13, and 19). The change in dielectric constant caused by this slight temperature variations can be neglected [34].

5) Impact of Earthquake: After excluding the possible factors as mentioned above, we need to give a further analysis to explain the changes in BC residual in the water bodies of the Yellow River and lakes around the epicenter on May 26, 2021. For the BC changes in the Yellow River on May 26, 2021 [see Fig. 3(b) and (c)], the satellite optical images with high spatial resolution before and after the Madoi M 7.3 earthquake were used to compare the changes along the river. The comparison of 2-m resolution GF-1 image on May 3, 2021 [see Fig. 9(a)] and the 1-m resolution GF-2 image on June 5, 2021 [see Fig. 9(b)] clearly shows the increased discharge in segments A and B (cyan arrows in Fig. 9). For segment C, two contrasting colors (dark and light) have been observed on June 5, 2021 as the result of the Heihe River (clearer than the Yellow River) with the increased stream flow into the Yellow River [red arrows in subfigures of Fig. 9(a) and (b)]. The increased stream flows in both Yellow River and Heihe River are due to the melting of glacier and permafrost that is associated with the increased temperature (see Fig. 6) and soil moisture (Fig. 5 and [32]) after the main event. Although the increase in stream flow is a coseismic or postseismic response [35], [36], we do not have any observation to support this. We have seen coseismic water level in water well in the surrounding region of epicenter, but we are not showing this in the present study (personal communication with Anhua He). High BC positive residual has been observed in segment B on May 26, 2021, while low residual has been observed in segments A and C [see Fig. 9(c)]. Such variations cannot be attributed to the ice in the river since we did not find any ice in the mainstream of the Yellow River from the satellite image (GF-1 image of May 3, 2021) [see Fig. 9(a)]. The increase in BC was only observed in segment B [see Fig. 9(c)], although the increased stream flows have been observed in all segments [see Fig. 9(b)]. The increased BC in segment B could be due to the discharge of sediments due to the shaking and liquefactions caused by the main earthquake, which is likely to enhance the turbidity in water bodies in the epicentral areas. A large number of liquefaction patches have been observed at many places in the river valley around the epicentral region from the field investigation after the earthquake occurrence [4]. In segment C, few BC increases have been observed, which may be due to the limited spatial resolution of Sentinel-1 data that difficult to map in the image. An increase in BC on the land within the epicentral region was found in the data of June 1, 2021 [see Fig. 3(d) and (e)]. High BC values and increased soil moisture have been observed, while no high precipitation was observed during this period, which could be attributed to the liquefactions along the river with the close distance to the seismogenic fault [4], [5].

The spatial distribution of the lakes with increased BC on May 26, 2021 [see Fig. 4(d)] shows obvious dependence on epicenter and coseismic surface rupture, and we speculated that it should be associated with the Madoi M 7.3 earthquake event. Furthermore, we obtained the longitudinal distribution of the high BC residual rank (greater than 5 dB) on the days showing obvious variations in Fig. 4. A contrast longitudinal distribution has been observed on the preevent days (April 14 and 26, May 8, 2021) compared to five days after the main earthquake event (May 26, 2021) (see Fig. 10). The higher BC positive residuals were observed on April 14 and 26, 2021 due to ice-covered most of the lakes in this region. The decline in BC residual value on May 8, 2021 can be attributed to the majority of lake ice melting with the temperature rising (5.1 °C on May 8, 2021 as shown in Fig. 6). We observed that the lakes with high BC positive residuals (most of them greater than 10 dB) during the preevent days distributed across the whole region as suggested by the longitudinal distribution (see Fig. 10). However, the lakes with high BC positive residuals on May 26, 2021 are found to be distributed on the eastern area of the epicenter, and with lower residual value (less than 10 dB) compared with the preevent days (red bars in Fig. 10). The lakes with increased BC are almost located on the east of epicenter, which is related to the focal mechanism. Wang *et al.* [3] proposed the more favorable conditions for rupture on the eastern part of the fault compared to the western parts due to vertical fault dip after the detailed analysis of focal mechanisms of earthquake sequence. InSAR and GPS observations show that the maximum coseismic slip is located in the eastern part of the main fault [6], [37].

To explain the possible reason for the water bodies with distinct increased BC on May 26, 2021 [see Fig. 4(d)], we further selected 14 locations (100 m × 100 m size for each) to analyze the BC changes before and after the main earthquake event. The locations P1–P7 with high BC residual rank (>5 dB) on May 26, 2021 are located in the north side of the surface rupture. For comparison, the locations P8–P14 are considered on the same side and with lower BC residual rank. According to the area affected by sand liquefaction

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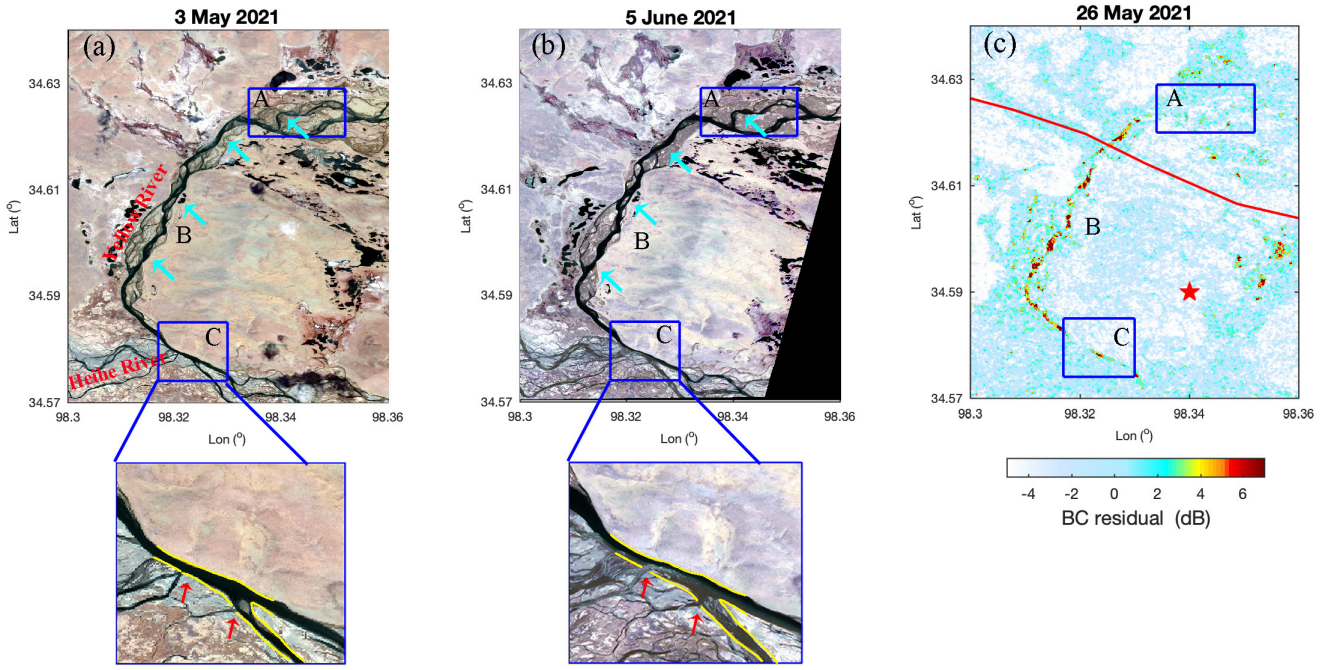


Fig. 9. Variation of river flood plains of the Yellow River prior and after the Madoi M 7.3 earthquake. (a) Image with a spatial resolution of 2 m observed by GF-1 satellite on May 3, 2021. (b) Image with a spatial resolution of 1 m observed by GF-2 satellite on June 5, 2021. (c) Changes in Sentinel-1 BC residuals at VV polarization on May 26, 2021 (compared to May 20, 2021). Yellow lines in the subfigures outline the river channel.

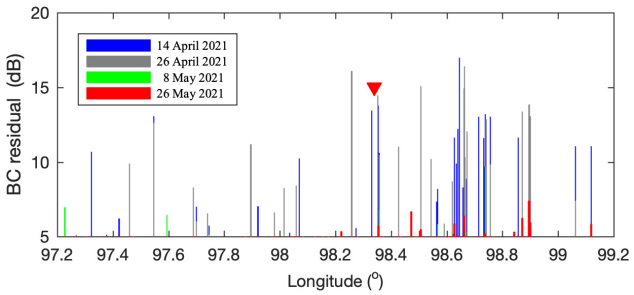


Fig. 10. Longitudinal distribution of the locations with BC residual greater than 5 dB (compared with May 20, 2021) in the 83 lakes. Red triangle indicates the longitude of the epicenter.

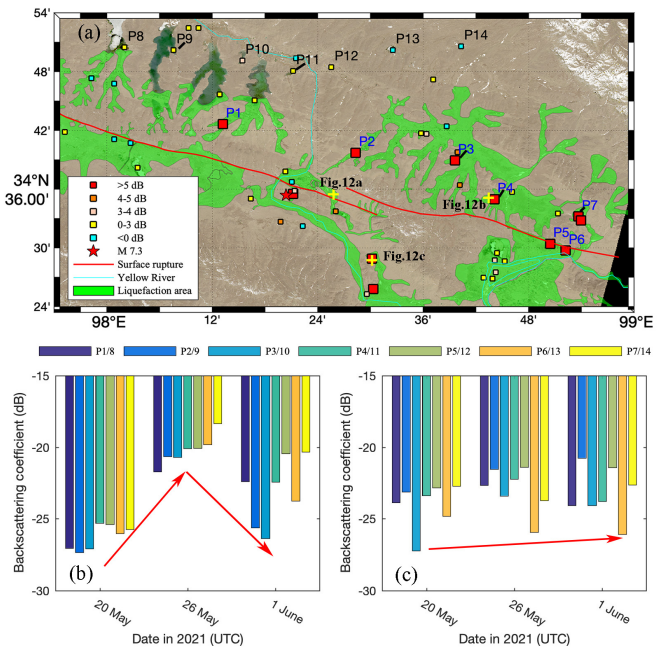


Fig. 11. Comparison of BCs at different locations shows earthquake and without earthquake-affected effect. (a) Location of the 14 selected points. (b) BC changes on the earthquake-affected points P1–P7. (c) BC changes on the without earthquake-affected points P8–P14. Base map is Landsat 8 image acquired on April 29, 2021. Yellow plus signs show the location of UAV photographs in Fig. 12.

from field observations [8] [green region in Fig. 11(a)], the locations P1–P7 are just within the liquefaction-affected area and P8–P14 are outside of the area, which clearly shows that liquefaction could be the main factor for the change in BC values due to the enhanced turbidity in those lakes caused by the main earthquake event. Fig. 11(b) and (c) clearly shows the different changes at the locations P1–P7 and P8–P14 before and after the main earthquake event. Xu *et al.* [8] observed the turbid lake water caused by sand liquefaction in the epicentral region after four days of the main event in the field investigation. The images from unmanned aerial vehicle (UAV) observation in the immediate field survey after the main earthquake event show the turbid water in the lake and liquefaction pockets along the lake in the epicentral region (see Fig. 12). Red circles in Fig. 12 show the turbid lake water on May 26 and 27, 2021. Yellow plus symbols in Fig. 11(a) show the location of observations. The turbid lake water observed by UAV on May 27, 2021 [see Fig. 12(b)] is close to the location with enhanced BC from Sentinel-1 data on May 26, 2021 [see Fig. 11(a)]. In the field investigation, we clearly

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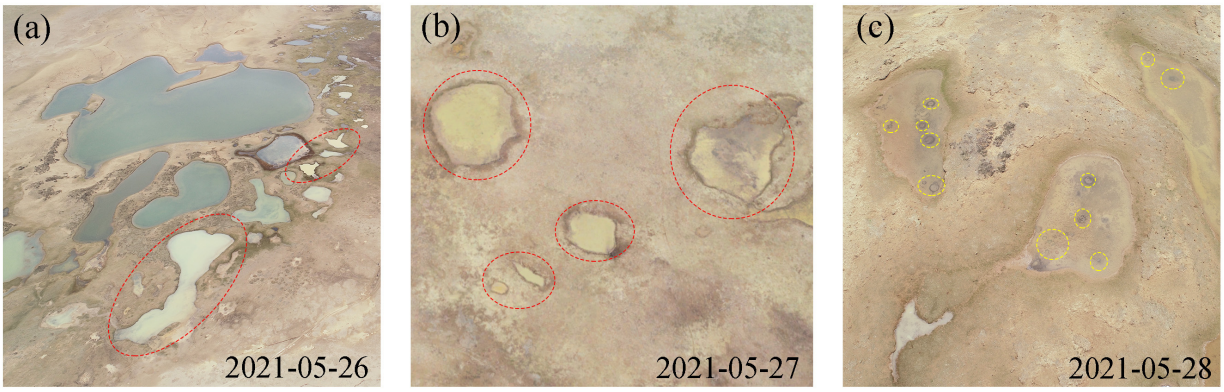


Fig. 12. Typical UAV photographs of (a) and (b) turbid lake water and (c) liquefaction pockets at the bottom of the lake observed after the M7.3 Madoi earthquake. Red circles show the turbid lake water and yellow circles show liquefaction pockets.

TABLE II
SENTINEL-1 DATA IN THE YEARS OF 2017–2020 FOR COMPARISON

Year	2017	2018	2019	2020
Reference day	17 May	12 May	19 May	13 May
Target day	29 May	24 May	31 May	25 May

observed liquefaction pockets in the bottom of dry lake [see Fig. 12(c)], which indirectly provides evidence of turbidity in the epicentral region due to lateral motion. In addition, the ground shaking caused by the main event could also disturb the bottom sediments in these water bodies that generate the suspended sediments in the water column showing changes in BC.

V. DISCUSSION

We have analyzed the BS variations in the water bodies (river and lakes) around the epicenter of Madoi earthquake in 2021. By excluding various factors such as precipitation, water/ice cover, surface roughness, dielectric constant, and combining with the distribution of liquefaction in the field investigation, we proposed that the high positive BC residual values observed on May 26, 2021 [see Figs. 3(b) and (c) and 4(d)] are related to the strong earthquake event. Furthermore, we analyzed the Sentinel-1 BC values at the same locations during the same period for the years 2017–2020. Due to the limitation of revisit period for polar-orbiting satellites, it was impossible to obtain BC value on the same day in the different years. We checked the available Sentinel-1 data and used the day in late May as the targeted day and the day in mid-May as the reference day as shown in Table II for comparison. The BC changes in the Yellow River and the 83 lakes have been analyzed using the same method in Section IV.

First, we obtain the BC residual distributions in Box A [location shown in Fig. 3(a)] to see the BC changes in the Yellow River (see Fig. 13). No high BC positive residual values similar to the observation on May 26, 2021 [see Fig. 9(c)] have been observed in the water bodies along the Yellow River on

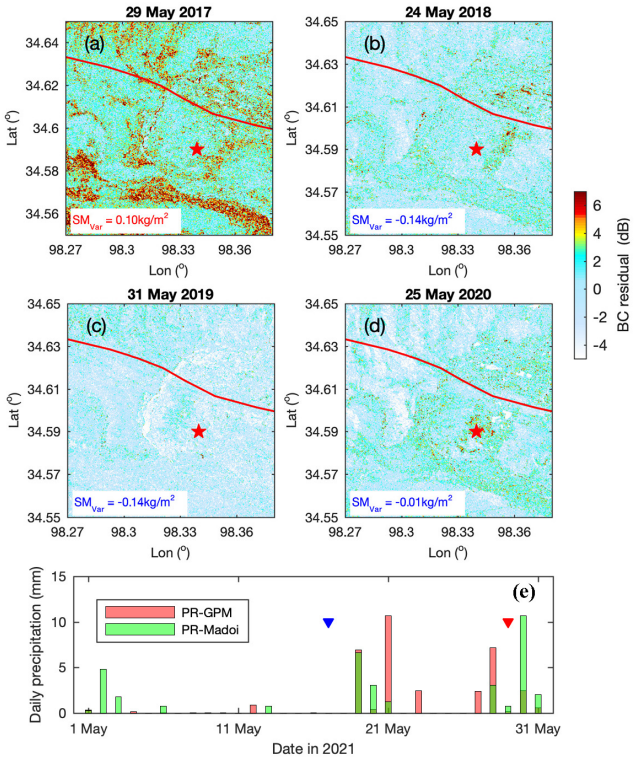


Fig. 13. BC changes in Box A in late May from 2017 to 2020. The legend is the same as Fig. 3(b). SM_{Var} in (a)–(d) indicates the difference in soil moisture between the target day and the reference day in this region. (e) Precipitation variation in May 2017 from satellite and ground observations. Blue and red triangles indicate the reference day and target day, respectively.

the target days from 2017 to 2020. However, we observed high residuals widely distributed on the land surface on the target day in 2017 (May 29), although not observed in the Yellow River [see Fig. 13(a)]. The GLDAS daily soil moisture data during this period were used to analyze the increased land BC. The soil moisture variations (SM_{Var}) of the target day relative to the reference were computed. The increased soil moisture [see Fig. 13(a)] has been observed in 2017 instead of negative values in other years [see Fig. 13(b)–(d)]. The precipitation

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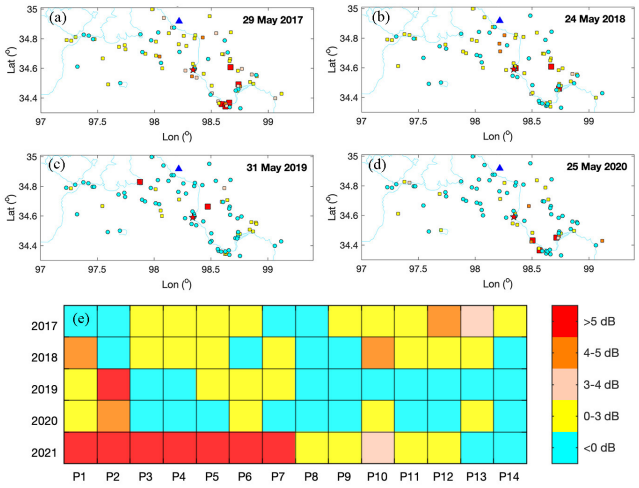


Fig. 14. Comparison of BC residual in (a)–(d) 83 lakes and (e) 14 locations during the same period for different years. Legend for (a)–(d) is the same as Fig. 4. The locations for P1–P14 in (e) are shown in Fig. 11(a).

data from GPM and Madoi station in May 2017 clearly show the increased soil moisture related to the precipitation during this time. The high precipitation value before the target day (May 29, 2017), the lack of precipitation on the reference day (May 17, 2017), and its earlier days have been observed [see Fig. 13(e)].

The four-year BC residuals in late May for the 83 selected lakes around the epicenter [see Fig. 14(a)–(d)] clearly show only few lakes with high BS positive residuals in periods 2017–2020 and the distribution pattern is different from the same period in 2021 that is distributed around the epicenter and along the surface rupture zone, as shown in Fig. 4(d). We also computed the BC residual ranks on the 14 points (P1–P14) on the target days from 2007 to 2020 and compared with the result in 2021 [see Fig. 14(e)]. The BC residuals of P1–P7 in 2021 show the distinct behavior compared with other years. For P8–P14 locations that are not affected by the earthquake, no high BC residual rank was observed on the target day during the periods 2017–2020 [see Fig. 14(e)].

Based on the analysis of various possible impact factors and comparison of data during the same period for different years, we can say that the BC changes on May 26, 2021 in the water bodies around the epicentral region should be associated with the Madoi M 7.3 earthquake event. Due to the limitation of observation conditions at high altitude (depopulated zone and inaccessible for most areas), it is impossible to obtain the total suspended sediments data of those water bodies that showing BC changes in the present study. Nevertheless, the turbid lakes and the liquefaction pockets at the bottom of the lake observed from the field investigation provide strong evidence to support the enhanced BC induced by the main earthquake event.

VI. Conclusion

In the present study, the enhanced turbidity in water bodies (river and lakes) associated with the 2021 Madoi M 7.3 earthquake has been analyzed using satellite and ground observation data. In the epicentral region, water turbidity shows changes, which is reflected from SAR BC. The combined satellite and ground observations clearly show pronounced changes in BCs at many locations in lake water and also along the Yellow River due to the strong changes in water quality associated with the Madoi M 7.3 earthquake. The observation from immediate field investigation shows liquefaction at many locations in the epicentral region. At the liquefaction pockets and at different locations in the lake water, we observed changes in BC associated with the seismic wave propagation affecting turbidity of lake water in the epicentral region. It is difficult to map increased soil moisture and small locations of liquefaction using SAR BC due to its limited spatial resolution of Sentinel-1 data, although the liquefactions at many locations may have enhanced soil moisture in the epicentral region.

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